Smart Agriculture Database Management System

Project Report

Project Title	Smart Agriculture Database Management System
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Database Engine	MySQL
Tools Used	MySQL Workbench, SQL

1. Introduction

Agriculture is one of the most vital sectors of the economy, and managing information about farmers, their farms, crops, and yields efficiently is essential for planning and decision-making. The **Smart Agriculture Database Management System** is a MySQL-based database project designed to organize and manage this information in a structured, reliable, and query-friendly way.

The system models the real-world relationship between farmers and the farms they own, the crops grown on those farms, and the yield produced from each crop, while also maintaining an automatic audit trail of new yield entries.

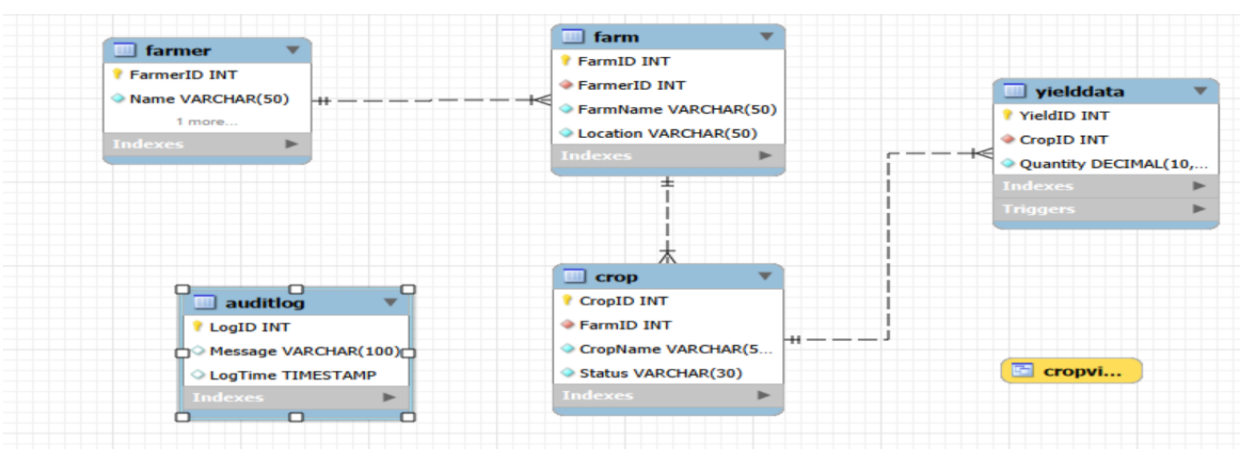
2. Objectives

- Design a normalized relational schema for agricultural data.
- Demonstrate practical use of primary keys, foreign keys, and referential integrity.
- Implement CRUD operations, aggregate functions, and multi-table joins.
- Build a view for simplified reporting of crop data.
- Implement a stored procedure to generate crop reports on demand.
- Implement a trigger to automatically log new yield entries for auditing.

3. System Overview

The database consists of five interrelated tables: **Farmer**, **Farm**, **Crop**, **YieldData**, and **AuditLog**. Each farmer can own multiple farms, each farm can grow multiple crops, and each crop can have multiple yield records recorded over time. The relationships form a one-to-many chain: Farmer → Farm → Crop → YieldData.

3.1 Entity-Relationship Diagram



4. Database Tables

Table	Purpose
Farmer	Stores farmer name and contact information
Farm	Stores farms owned by each farmer, with location
Crop	Stores crops grown on each farm and their growth status
YieldData	Stores the yield quantity produced for each crop
AuditLog	Automatically records a message whenever new yield data is inserted

5. Key Features Implemented

- **CRUD Operations** - INSERT, SELECT, UPDATE, and DELETE demonstrated across tables.
- **Aggregate Functions** - COUNT, SUM, AVG, MAX, and MIN used on yield and farmer data.
- **INNER JOIN** - combining Farmer, Farm, and Crop tables to produce consolidated reports.
- **VIEW (CropView)** - a simplified, reusable view exposing only crop name and status.
- **Stored Procedure (CropReport)** - returns a formatted list of all crops on demand.
- **Trigger (YieldLog)** - automatically inserts an audit entry after every new yield record.

6. Technologies Used

MySQL, MySQL Workbench, and standard SQL (DDL, DML, DQL, Views, Stored Procedures, and Triggers).

7. Conclusion

The Smart Agriculture Database Management System successfully demonstrates the design and implementation of a normalized relational database with real-world applicability. Through the use of joins, views, stored procedures, and triggers, the project shows a practical, end-to-end understanding of relational database concepts that can be extended further with a front-end application or reporting dashboard.